

3. 数值比较

$$\text{error} = \|X_{\text{程序}} - X_{\text{Abaqus}}\|_2 / \|X_{\text{Abaqus}}\|_2$$

[节点位移对比]

		Abaqus	程序	全场误差
UMAG	Max	1.776078e+01	1.776078e+01	1.953077e-08
	Min	8.151400e-34	0.000000e+00	
U1	Max	1.110547e+01	1.110547e+01	2.470603e-08
	Min	-6.530023e-01	-6.530023e-01	
U2	Max	4.370890e+00	4.370890e+00	2.578498e-08
	Min	-1.415522e+01	-1.415522e+01	
U3	Max	2.678854e+00	2.678854e+00	2.530059e-08
	Min	-2.458777e+00	-2.458777e+00	

[原始单元节点应力对比]

		Abaqus	程序	全场误差
Mises	Max	1.781279e+04	1.781279e+04	3.936133e-08
	Min	5.088124e+00	5.088125e+00	
S11	Max	4.920850e+03	4.920850e+03	2.529972e-08
	Min	-1.327367e+04	-1.327367e+04	
S22	Max	4.976667e+03	4.976667e+03	2.527820e-08
	Min	-1.358818e+04	-1.358818e+04	
S33	Max	3.002371e+03	3.002371e+03	2.543304e-08
	Min	-8.844631e+03	-8.844631e+03	
S12	Max	5.506032e+03	5.506032e+03	2.551107e-08
	Min	-2.228924e+03	-2.228924e+03	
S13	Max	3.206939e+03	3.206939e+03	2.530146e-08
	Min	-5.272838e+03	-5.272839e+03	
S23	Max	6.484059e+03	6.484059e+03	2.541514e-08
	Min	-4.017857e+03	-4.017857e+03	

数据说明：节点位移共 31,167 条，原始单元节点应力共 533,292 条。程序结果与 Abaqus 结果分别取自 complicate_cae_model 和 complicate_cae_model_odb，两组数据按原始行序和 node_id 对齐。